Hivemind DHT control plane: timing reference

Writes and reads on the averaging-availability key, against the protocol's own timeouts. Measurements are loopback on one machine and are therefore **floors**. hivemind 1.2.0.dev0.

1 WHAT THE PROTOCOL ALLOWS

PARAMETER	DEFAULT	SOURCE	WHAT IT BOUNDS
wait_timeout	**3.0 s**	`dht/node.py:108`	One DHT RPC: rpc_ping, rpc_store, rpc_find. Past this the peer is deemed lost.
blacklist_time	**5.0 s**	`dht/node.py:119`	How long a non-responsive peer is excluded, multiplied by backoff_rate each time.
request_timeout	**3.0 s**	`averaging/averager.py:122`	Waiting for a leader's reply after asking to join. Must be < min_matchmaking_time.
min_matchmaking_time	**5.0 s**	`averaging/averager.py:121`	Advertisement lifetime: expiration = get_dht_time() + this. Also the leadership priority.
num_replicas	**5**	`dht/node.py:138`	Peers asked to store a record. The cost of custody.
bucket_size	**20**	`dht/node.py:104`	Peers seeding a lookup traversal. The cost of controlling what a victim sees.

Advertisement lifetime and leadership priority are **the same field** (`key_manager.py:60`, read at `:88`), so lowering the deadline to win also shortens the advertisement. That only bites at second scale: expirations are absolute timestamps near 1.79e9, where **one float step is 2.4e-7 s**. Undercutting by `nextafter` wins outright for 240 ns of lifetime, so the coupling is not the defence it appears to be.

2 COST OF A WRITE AND A READ, HEALTHY SWARM

SWARM	WRITE MEDIAN (MS)	WRITE RANGE	READ MEDIAN (MS)	READ RANGE	SAMPLES
8 peers	**3.2**	2.5–6.5	**2.5**	2.1–3.1	30
16 peers	**5.5**	4.8–6.5	**4.8**	4.1–5.8	30
32 peers	**6.7**	5.8–9.1	**5.7**	5.2–8.4	30

`declare_averager` writes to `num_replicas` peers; `get_averagers` reads with `latest=True`, which sets `sufficient_expiration_time` to infinity (`node.py:542`) so the traversal **cannot stop early on a good answer**. 30 samples per row.

3 WHAT ONE SLOW PEER DOES TO A READ (8 PEERS, ONE ATTACKER)

ATTACKER DELAY	VICTIM READ MEDIAN (MS)	OUTCOME
0 s	**2.5**	no attacker present
0.5 s	**506.0**	attacker answers late, never blacklisted
1.5 s	**1506.5**	attacker answers late, never blacklisted
2.9 s	**2912.8**	just under wait_timeout, maximum damage, still undetected
3.5 s	**2.7**	past wait_timeout, RPC times out, attacker IS blacklisted, attack fails

The attacker sets the victim's whole read time, because the traversal waits for the slowest responder it contacts. More slow peers change nothing: 1, 2 and 4 all give ~2.91 s.

4 WHETHER A NEW ADVERTISEMENT IS EVER SEEN (OBSERVED / TRIALS)

ADVERTISEMENT LIFETIME	NO DELAY	ATTACKER DELAYS 0.5 S	ATTACKER DELAYS 1.5 S	ATTACKER DELAYS 2.9 S
1 s	3/3	2/3	**0/3**	**0/3**
2 s	3/3	3/3	**0/3**	**0/3**
3 s	3/3	3/3	3/3	**0/3**
5 s (Hivemind default)	3/3	3/3	3/3	**0/3**
8 s	3/3	3/3	3/3	3/3

A stretched read outlives the advertisement it fetched: entries are dropped by `_remove_outdated()` on read-out (`timed_storage.py:96`). The invisibility threshold is about twice the attacker's delay, a number the attacker chooses.

5 TIMINGS THAT DECIDE THE LEADER ELECTION

QUANTITY	VALUE	CONSEQUENCE
advertisement lifetime	**5.0 s**	Set by `min_matchmaking_time`; also the election priority.
one float step at these timestamps	**2.4e-7 s**	The cost of undercutting the earliest deadline by `nextafter`.
blind undercut, §4.3 strategy	**1.0 s declared**	Wins while alive, but the advertisement survives about 20 % of the window.
attacker read then declare budget	**≥ 2 s**	Measured tolerance between reading the key and declaring, at a 5 s window.

Measured over 32-peer swarms, 2 repeats: an attacker undercutting by one float step was followed by **100 % of honest peers** while holding the lowest `peer_id` in **0 of 40** samples , so the win came from the deadline, not from identity. Scope: the election is computed by applying hivemind's own comparison rule to real DHT records, not observed in live matchmaking.

> **Reading these numbers.** Loopback removes network latency, so sections 2 to 3 are lower bounds and the attack margins are upper bounds. The claim to check first is section 4: an advertisement at hivemind's shipped 5 s lifetime was never observed by a victim whose reads were stretched to 2.9 s, across three trials.